Lecture 2: Learning Activities

Problem 1: Find candidate keys in relation AD CAMPAIGN MIX?

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Given FDs:

A→B
A→C
A→D
A→E
A→F
E→F
G→H
G→I
AG→J

Lossless Join Decomposition

Problem 2a: Find R1 ⋈ R2?

R(A,B,C)

A	B	C
1	1	1
2	1	2
3	2	1
4	3	2

R1(A,B)

A	B
1	1
2	1
3	2
4	3

R2(B,C)

B	C
1	1
1	2
2	1
3	2

⋈ represents natural join.

Problem 2b: Find R1 ⋈ R2?

R(A,B,C)

A	B	C
1	1	1
2	1	2
3	2	1
4	3	2

R1(A,B)

A	B
1	1
2	1
3	2
4	3

R2(A,C)

A	C
1	1
2	2
3	1
4	2

Problem 3: Consider a relation R(A, B, C, D, E, H) on which the following functional dependencies hold:

A→ B

BC→ D

E→ C

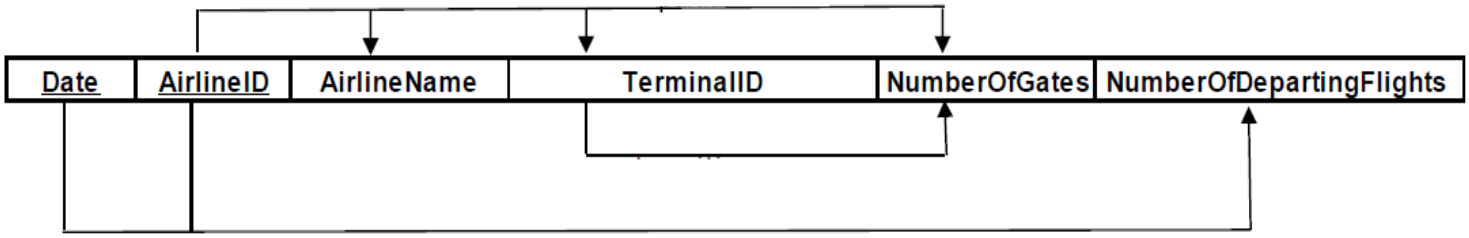
D→ A

What are the candidate keys of R?

(a) AE, BE
(b) AE, BE, DE
(c) AEH, BEH, BCH
(d) AEH, BEH, DEH

Problem 4: Find the types of FD in Relation Airport?

Graphical representation of FDs of the relation AIRPORT



Primary Key= {Date, AirlineID}

Problem 5: Decompose the relation AIRPORT such that it must satisfy second normal form (2NF)?